Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

mninadmnobo@gmail.com github.com/mninadmnobo linkedin.com/in/mninadmnobo
Portfolio (mninadmnobo.github.io) Google Scholar (M Ninad M Nobo) +880 1939-444451

Summary

CSE undergraduate at BUET focused on building intelligent systems at the intersection of large language models, software automation, and applied machine learning systems. Hands-on experience designing end-to-end AI-driven systems across mobile and backend platforms, with strong emphasis on reliability, scalability, and real-world deployment. Actively engaged in research on LLM-based software testing and medical AI, including dataset creation, evaluation pipelines, and trust-aware decision systems.

Education

Bangladesh University of Engineering and Technology (BUET) **2022 – 2026**
Bachelor of Science in Computer Science and Engineering (CSE) **CGPA: 3.60 / 4.00**
Relevant Coursework: Data Structures and Algorithms, Digital Logic Design, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Computer Graphics, Algorithm Engineering, Artificial Intelligence, Machine Learning
Research Interests: LLM Systems, Software Testing Automation, Applied Machine Learning, Medical AI, Computer Networks

Research Experience

- **AutoTestGenX: Multi-Agent LLM Framework for End-to-End Web Testing** [GitHub]
  Python • LLM Engineering • Multi-Agent Systems • Web Testing • Browser Automation *2025 – 2026*
  – Developed a multi-agent LLM framework for automated web test generation and execution from natural-language requirements.
  – Built benchmark datasets and evaluated generated test suites against engineer-validated ground-truth test suites.
  – Achieved 84.0% test scenario coverage and 90% error detection, outperforming zero-shot and few-shot prompting approaches.
- **MedCAR: Conflict-Aware Medical Reasoning System (Under Review)** [GitHub]
  Python • Deep Learning • Medical AI *2026*
  – Built a multi-model system for chest X-ray analysis integrating heterogeneous AI tools.
  – Developed a conflict resolution pipeline to reconcile contradictory outputs using semantic reasoning and confidence calibration.
  – Designed trust-aware decision framework with uncertainty-based abstention for reliable predictions.
- **Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization** [arXiv: 2602.14291]
  Python • Speech Processing • Machine Learning *2026*
  – Contributed to building a large-scale benchmark dataset for long-form Bangla ASR and speaker diarization in a multi-institution collaboration, developing data collection and annotation pipelines for real-world speech data.
  – Designed subtitle extraction and annotation workflows enabling evaluation with WER and DER.

Projects

- **MindTrace: AI-Powered Dementia Care Application** [GitHub | Feature | Infrastructure]
  Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure *2025*
  – Engineered an AI-driven mobile–backend system for dementia care with an Android frontend and scalable Spring Boot architecture, enabling real-time caregiver workflows and seamless system integration.
  – Designed accessibility-first UI/UX for cognitively impaired users, focusing on clarity, simplified interactions, and usability to enhance engagement and reduce cognitive load.
- **Gemma VetCare: AI-Assisted Farming Support System** [GitHub | Feature]
  Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB *2025*
  – Built an AI-assisted Android application for veterinary decision support, integrating backend services to deliver real-time recommendations for diagnosis and treatment workflows.
  – Designed RESTful APIs optimized for low-connectivity environments with efficient request handling and streamlined data flow.
- **Compiler Construction (Flex, Bison, 8086 Codegen)** [GitHub]
  C++17 • Flex • Bison/Yacc • Linux *2024*
  – Designed and implemented a full compiler pipeline for a C-like language, encompassing lexical analysis, parsing, semantic analysis, and intermediate code generation.
  – Generated optimized 8086 assembly code with symbol table management and basic optimizations.
- **Computer Graphics Pipeline Implementation** [GitHub]
  C++17 • OpenGL/GLUT • Rasterization • Ray Tracing *2025*
  – Implemented a complete graphics pipeline including transformations, clipping, rasterization with Z-buffering, and ray tracing.
  – Developed OpenGL-based rendering demos with camera systems supporting 6-DOF control and real-time interaction.

Technical Skills

Languages: C/C++, Python, Java, Kotlin, JavaScript, SQL
Backend & APIs: Spring Boot, Spring AI, Node.js, Express.js, REST APIs, Secure API Design
AI/ML: PyTorch, TensorFlow, Scikit-learn, LLM Applications
Databases: PostgreSQL, MongoDB, Redis
DevOps & Infrastructure: Linux, Git, Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, Firebase

Additional Information

Spoken Languages: Bangla (Native), English (Fluent), Hindi & Urdu (Conversational), Arabic (Reading)
Programming Profiles: NeetCode: mninadmnobo LeetCode: mninadmnobo
Research Profiles: ORCID: 0009-0006-2781-6693 ResearchGate: Mohammad Ninad Mahmud Nobo